

Grade 11 Physics — Practice Exam

Thai Mattayom 5 (OBEC Curriculum) - 40 Multiple Choice Questions

Name: _____ Class: _____ Date: _____

Instructions: Circle the letter of the correct answer for each question. Each question is worth 1 point (40 points total). $g = 10 \text{ m/s}^2$ unless otherwise noted.

Work, Energy, and Power

1. A person pushes a box with a force of 50 N through a displacement of 4 m in the direction of the force. How much work is done?

- (A) 12.5 J
- (B) 46 J
- (C) 54 J
- (D) 200 J

2. A force of 20 N acts at an angle of 60° to the direction of a 5 m displacement. What is the work done?

- (A) 100 J
- (B) 86.6 J
- (C) 50 J
- (D) 0 J

3. A waiter carries a tray horizontally at constant velocity across a room. What is the work done by the upward force from the waiter's hand on the tray?

- (A) Positive
- (B) Negative
- (C) Zero
- (D) Cannot be determined

4. A 2 kg object is lifted at constant velocity through a height of 3 m. Find the work done against gravity. ($g = 10 \text{ m/s}^2$)

- (A) 6 J
- (B) 20 J
- (C) 60 J
- (D) 600 J

5. Power is best defined as:

- (A) Work done $\times$ time
- (B) Work done $\div$ time
- (C) Force $\div$ time
- (D) Energy $\div$ distance

6. A motor lifts a 500 N load to a height of 10 m in 5 seconds. What is the power output of the motor (ignoring losses)?

- (A) 100 W
- (B) 250 W
- (C) 1000 W
- (D) 5000 W

Momentum and Collisions

7. Momentum is defined as the product of:

- (A) Mass and acceleration
- (B) Mass and velocity
- (C) Force and time
- (D) Force and displacement

8. A 1000 kg car traveling at 20 m/s collides with an identical stationary 1000 kg car, and the two lock together. Find their common velocity after the collision.

- (A) 5 m/s
- (B) 10 m/s
- (C) 15 m/s
- (D) 20 m/s

9. In a perfectly elastic collision between two objects:

- (A) Only momentum is conserved
- (B) Only kinetic energy is conserved
- (C) Both momentum and kinetic energy are conserved
- (D) Neither quantity is conserved

10. A 0.5 kg ball moving at 6 m/s hits a wall and bounces straight back with the same speed. What is the magnitude of the change in momentum?

- (A) 0 kg·m/s
- (B) 3 kg·m/s
- (C) 6 kg·m/s
- (D) 12 kg·m/s

11. Impulse delivered to an object is equal to its:

- (A) Change in kinetic energy
- (B) Change in momentum
- (C) Change in velocity
- (D) Force divided by time

12. A 0.02 kg bullet is fired from a 5 kg gun initially at rest. If the bullet leaves with a velocity of 400 m/s, what is the recoil speed of the gun?

- (A) 0.8 m/s
- (B) 1.6 m/s
- (C) 3.2 m/s
- (D) 8 m/s

Rotational Motion and Equilibrium

13. The torque produced by a force F applied at a perpendicular distance r from a pivot is given by:

- (A) $\tau = F / r$
- (B) $\tau = F \times r$
- (C) $\tau = F + r$
- (D) $\tau = F^2 \times r$

14. Which condition must be satisfied for a rigid body to be in complete mechanical equilibrium?

- (A) Net force = 0 only
 - (B) Net torque = 0 only
 - (C) Both net force = 0 and net torque = 0
 - (D) Net force equals net torque
15. A wrench is used to turn a bolt. Applying the same force but at a longer distance from the bolt will:
- (A) Decrease the torque
 - (B) Increase the torque
 - (C) Have no effect on the torque
 - (D) Reverse the direction of rotation
16. The moment of inertia of an object depends on:
- (A) Mass only
 - (B) Mass and how it is distributed relative to the axis of rotation
 - (C) Angular velocity only
 - (D) Torque only
17. A force of 15 N is applied perpendicular to a wrench at a distance of 0.2 m from the bolt. What torque is produced?
- (A) 0.75 N·m
 - (B) 3 N·m
 - (C) 7.5 N·m
 - (D) 30 N·m
18. Two children sit on a seesaw. A 30 kg child sits 2 m from the pivot. How far from the pivot must a 40 kg child sit on the other side for balance?
- (A) 1.2 m
 - (B) 1.5 m
 - (C) 2.0 m
 - (D) 2.67 m

Gravitation

19. According to Newton's law of universal gravitation, the gravitational force between two masses is:
- (A) Directly proportional to the product of the masses and inversely proportional to the square of the distance between them
 - (B) Directly proportional to the distance between them
 - (C) Inversely proportional to the product of the masses
 - (D) Independent of mass and distance
20. If the distance between two objects is doubled, the gravitational force between them becomes:
- (A) Twice as strong
 - (B) Four times as strong
 - (C) One-half as strong
 - (D) One-quarter as strong
21. The acceleration due to gravity on the surface of a planet depends on:
- (A) The mass of the planet only
 - (B) The radius of the planet only
 - (C) Both the mass and radius of the planet

- (D) The mass of the object placed on it
22. A satellite orbits Earth in a circular path. The gravitational force acting on it serves as:
- (A) The force that slows it down
 - (B) The centripetal force keeping it in orbit
 - (C) A force with no effect on its motion
 - (D) The force that pushes it away from Earth
23. Weightlessness experienced by astronauts in an orbiting spacecraft occurs because:
- (A) There is no gravity in space
 - (B) The spacecraft and astronauts are in continuous free fall together
 - (C) The astronauts have zero mass
 - (D) Gravity acts only on the spacecraft, not the astronauts
24. If the mass of Earth were to double while its radius stayed the same, the gravitational acceleration at its surface would:
- (A) Stay the same
 - (B) Double
 - (C) Quadruple
 - (D) Reduce by half

Periodic Motion (SHM)

25. Simple harmonic motion (SHM) is characterized by a restoring force that is:
- (A) Constant in magnitude and direction
 - (B) Directly proportional to displacement and directed toward the equilibrium position
 - (C) Directly proportional to velocity
 - (D) Independent of displacement
26. At the equilibrium position of an object undergoing SHM:
- (A) Velocity is maximum and acceleration is zero
 - (B) Velocity is zero and acceleration is maximum
 - (C) Both velocity and acceleration are zero
 - (D) Both velocity and acceleration are maximum
27. The period of a simple pendulum depends on:
- (A) The mass of the bob and the length of the string
 - (B) The length of the string and gravitational acceleration
 - (C) The amplitude of swing only
 - (D) The mass of the bob only
28. A simple pendulum has a length of 1 m. Using $g = 10 \text{ m/s}^2$ and $\pi^2 \approx 10$, find its period.
- (A) 1 s
 - (B) 2 s
 - (C) π s
 - (D) 4 s
29. If the length of a simple pendulum is quadrupled, its period will:
- (A) Double
 - (B) Quadruple
 - (C) Remain the same

- (D) Become half
30. For a mass–spring system in SHM, increasing the spring constant k (keeping mass constant) will:
- (A) Increase the period
 - (B) Decrease the period
 - (C) Not affect the period
 - (D) Make the motion non-periodic
31. The amplitude of an oscillating system in SHM represents:
- (A) The maximum displacement from equilibrium
 - (B) The total distance traveled in one period
 - (C) The time for one complete oscillation
 - (D) The maximum velocity of the object
32. A mass on a spring completes 10 oscillations in 5 seconds. What are its period and frequency?
- (A) $T = 0.5 \text{ s}$, $f = 2 \text{ Hz}$
 - (B) $T = 2 \text{ s}$, $f = 0.5 \text{ Hz}$
 - (C) $T = 5 \text{ s}$, $f = 10 \text{ Hz}$
 - (D) $T = 0.2 \text{ s}$, $f = 5 \text{ Hz}$

Mechanical Waves and Sound

33. In a transverse wave, the particles of the medium vibrate:
- (A) Parallel to the direction of wave propagation
 - (B) Perpendicular to the direction of wave propagation
 - (C) In circles around a fixed point
 - (D) They do not vibrate at all
34. Sound waves traveling through air are classified as:
- (A) Transverse waves
 - (B) Longitudinal waves
 - (C) Electromagnetic waves
 - (D) Standing waves only
35. The speed of a wave is related to its wavelength (λ) and frequency (f) by the equation:
- (A) $v = \lambda/f$
 - (B) $v = \lambda f$
 - (C) $v = f/\lambda$
 - (D) $v = \lambda + f$
36. A wave has a frequency of 250 Hz and a wavelength of 1.4 m. What is its speed?
- (A) 178.6 m/s
 - (B) 251.4 m/s
 - (C) 350 m/s
 - (D) 3500 m/s
37. The pitch of a sound is most closely related to its:
- (A) Amplitude
 - (B) Frequency
 - (C) Wave speed

(D) Wavelength, independent of medium

38. As a sound wave travels from air into water, which property remains unchanged?

- (A) Speed
- (B) Wavelength
- (C) Frequency
- (D) Amplitude

39. The loudness of a sound is primarily determined by the wave's:

- (A) Frequency
- (B) Amplitude
- (C) Wavelength
- (D) Speed

40. Two sound waves interfere constructively when they meet:

- (A) In phase, with crest aligning with crest
- (B) Out of phase, with crest aligning with trough
- (C) With different frequencies only
- (D) With different amplitudes only